Some tips for writing an AI/ML paper

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Writing is an art**.** Overall, it is extremely helpful if you read lots of good papers and understand how they are written. In addition, these points might be useful.

- [**Principle 1 - Write a holistic and CONSISTENT narrative**]: Writing abstract->intro->main body is like looking at the same item from different scopes. Abstract is like a telescope, Intro is like bare eyes, main body is like a microscope, but you are always looking at the **SAME ITEM**. So, everything needs to be consistent throughout the entire paper, and you reiterate a few times for these points. E.g., Related Work should illustrate the gap of existing works, Methodology should support your claim about the novelty and technical contribution, and Experiments should support the claims about effectiveness.

- [**Principle 2 - Researchers help researchers**]: We should do our best to minimize the readers'/reviewers' effort in understanding the paper. You make reviewers happy; reviewers make you happy.
    - No overly long sentences/paragraphs, no mumbling
    - Be upfront, no cast wind-up (施法前摇)
    - Be logical
    - Revise your paper in the shoes of the readers/reviewers

- [**Abstract**] Abstract usually has four parts: 1) The general background on what theme our paper is about. 2) What is the gap in existing works, and what challenges do we have? 3) What method do we propose, and what's new in our method? 4) What do theoretical and empirical results look like?

- [**Introduction**] This is an expanded version of the Abstract. The rest of the paper is a further expansion of the Introduction.

- [**Related work**] The goal is to 1) show that we have a good understanding of the literature, and 2) show that there is a gap in existing work which we aim to fill in
  - How to efficiently do a **literature review** (for AI/ML papers)?
  - 1) Identify a list of keywords
  - 2) Use the keywords to search for the most recent papers (e.g., last 2 years) from top ML/AI venues (ICLR, ICML, NeurIPS, AAAI, RLC, etc.). Get more from the “Related Work” sections in these papers. These are papers that we MUST include.
  - 3) Expand the search in Google Scholar and include more (less recent, other venues) when appropriate.
  - 4) Create a list of all papers that you found, e.g., using Google Sheets
  - 5) Skip through the abstracts and the paper overall to gauge the quality and relevance. Pick the top K papers to read more thoroughly. These are the papers that we may need to compare with.

- [**Reference styles**] BibTeX Guidelines for Conference and Journal Papers
  - **Journal articles**: @article
    - Required: author, title, journal, year
    - Optional: volume, number, pages, doi
  - **Conference papers**: @inproceedings
    - Required: author, title, booktitle, year
    - Optional: pages
  - **Rule of thumb:**
    - Prefer peer-reviewed versions over arXiv whenever applicable
    - Do not include location (city, country) or publisher
    - Use URL is fine for arxiv or openreview, but remove hyperlinks

- [**Preliminary (optional) + Methodology**]
  - Use visuals to explain the method whenever appropriate.
  - Start with “WHY”, and then “WHAT” or “HOW”

- It is about OUR METHOD: The ultimate goal is always to highlight what is new in our method.
- Tell a story, not a list of bullets
- Be clear, concise, and consistent
- Always explain your notation the first time you introduce it
- Justify choices: Always explain why specific methods/models are chosen
- Use visuals to illustrate model architectures whenever possible
- Use algorithmic blocks for methodological pipelines whenever possible

- [**Equations**]:
    - Needs to be consistent whether using a comma/period at the end
    - No space between text and equation, which saves space
    - References to equations should be Eq.\eqref{}:
      1) no space, 2) eqref not ref 3) first letter should be upper case

- [**Experimental results]:**
    - You should first have a **concrete and thorough plan** before you start your experiment pipeline (so that you are aware of the run time, etc). The plan should include exactly what **tables and figures** you want to put in the paper – you may adjust your plan later.
    - What to include in the experiment: (1) Research questions or hypotheses (2) Experiment setting: Datasets, model architectures, implementation details (hardware/software setup), evaluation metrics, baselines (3) Results and discussions
    - For the results and discussions: usually (1) a **core result**, (2) an **ablation study** session (if this is about a new algorithm), and (3) a set of **auxiliary results** (e.g., visualization, exploratory results, interpretation).

- [**Visuals/figures**] ML visuals Github

- [**Code** Submission] Most conference submissions encourage code submission, and because conferences are double-blindly reviewed, we need to generate an anonymous package.
    - One convenient way is to use [Anonymous GitHub].
    - You can also just upload the package